

**MIAE 215**  
**Programming for Mechanical and Industrial Engineers**  
Fall 2026 (section Y)

**Course Instructor:** Brandon W. Gordon  
**E-mail :** [brandon.gordon@concordia.ca](mailto:brandon.gordon@concordia.ca)

**Office Hours:** To be determined

**Tutorials:** Please see your class schedule for details. **There are no tutorials for the first THREE weeks.**  
Tutors:  
To be determined

**Labs:** Please see the upcoming lab schedule for details. **There are no labs for the first THREE weeks.**  
Lab Demonstrators:  
To be determined

**Course Calendar Description:**

Writing programs using assignment and sequences. Variables and types. Operators and expressions. Conditional and repetitive statements. Input and output. File access. Functions. Program structure and organization. Pointers and dynamic memory allocation. Introduction to classes and objects. Mechanical and industrial engineering applications. Lectures: three hours per week. Tutorial: one hour per week. Laboratory: one hour per week.

**Prerequisites:** MATH 204 (Cegep mathematics 105)  
**Co-requisites:** N/A

**Specific Knowledge and Skills Needed for this Course:**

**There are no labs and tutorials during the first three weeks. Students are instead expected to complete lessons #1 to #4 of the MIAE 215 online “mini course” (see link below) during the first week of the term and all mini course lessons by the end of the second week.**  
[https://users.encs.concordia.ca/~bwgordon/Getting\\_started\\_C++.html](https://users.encs.concordia.ca/~bwgordon/Getting_started_C++.html)

**Course materials**

**Recommended Textbook:** C++ A Beginner’s Guide, Second Edition". Herbert Schildt, McGraw-Hill Osborne. ISBN-13: 978-0-07-223215-8, ISBN-10: 0-07-223215-3. This book is available from Amazon in paperback or E-Reader form:

<https://www.amazon.ca/Beginners-Guide-Second-Herbert-Schildt/dp/0072232153/>

**Instructor’s lecture notes:** *will be posted in Moodle course management site*

**Lab Manual:** *will be posted in Moodle course management site*

**Software Use:** *covered in the online “mini course” described in the previous section*

## Grading

| Assessment Tool                                | Weight |
|------------------------------------------------|--------|
| Midterm                                        | 33%    |
| Final                                          | 50%    |
| Project (based on the Arduino labs)            | 7%     |
| Assignments (two hand-in assignments, 5% each) | 10%    |
| Total                                          | 100%   |

## Tentative Course Schedule

| Topics                                           | Week |
|--------------------------------------------------|------|
| Introduction to computer programming and C++     | 1    |
| Writing, compiling, and debugging programs       | 1    |
| Variable types                                   | 2    |
| Expressions and operators                        | 3    |
| Control statements                               | 4-5  |
| Arrays and C-strings                             | 6-7  |
| Arduino programming and mechatronic applications | 7-12 |
| Functions and modular programming                | 8-10 |
| File Input / Output (I/O)                        | 10   |
| Pointers                                         | 11   |
| Introduction to Python programming               | 12   |
| Computer algorithms and engineering applications | 5-12 |

## Lab Details

Students will learn how to program an Arduino microcontroller board for basic mechatronic applications. More details will be posted on Moodle later in the term.

## Engineering Tools

Programming software: C++ compiler

## Details on Assessment Tools:

### A. Term Project:

This course includes an individual Arduino project based on the labs. The details of the project will be given on Moodle later in the term.

## Other relevant information

Don't email me using Moodle -- use my email address indicated above.

ALL sections have a common final exam that is in-person

Lectures will be provided both in-person and online (hybrid approach) -- you can take lectures online or attend lectures in person.

Online lectures are provided in video files (like the mini-course) you can watch any time.

### Graduate Attributes:

The following is the list of graduate attributes (skills) that students use, learn and/or apply throughout the term.

| Graduate Attribute                    | Indicators                                                 | Level of Coverage | Assessment Results Reported |
|---------------------------------------|------------------------------------------------------------|-------------------|-----------------------------|
| <b>Knowledge base for engineering</b> | Knowledge base in a specific domain                        | Beginner          | Yes                         |
| <b>Problem Analysis</b>               | Problem Solving Analysis                                   | Beginner          | Yes                         |
| <b>Design</b>                         | Detailed design Validation and implementation              | Beginner          | Yes                         |
| <b>Use of engineering tools</b>       | Ability to use appropriate engineering tools and resources | Beginner          | Yes                         |

### Course Learning Outcomes (CLOs):

By the end of this semester, students are expected to master the following engineering concepts.

| Course Learning Outcome                                                                                                               | Related Graduate Attributes                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Understand the usage of C++ syntax including data types, operators, expressions, control statements, functions, arrays, and pointers. | <ul style="list-style-type: none"> <li>Knowledge base in a specific domain</li> </ul>                                                                     |
| Apply knowledge of C++ syntax in order to write computer programs.                                                                    | <ul style="list-style-type: none"> <li>Detailed design</li> <li>Validation and implementation</li> </ul>                                                  |
| Recognize the importance of functions for developing modular programs.                                                                | <ul style="list-style-type: none"> <li>Knowledge base in a specific domain</li> <li>Problem Solving</li> <li>Analysis</li> <li>Detailed design</li> </ul> |
| Analyze (read) C++ programs in order to predict the output of a given program.                                                        | <ul style="list-style-type: none"> <li>Knowledge base in a specific domain</li> <li>Problem Solving</li> <li>Analysis</li> </ul>                          |
| Design and write C++ programs to solve engineering applications including mechatronic applications.                                   | <ul style="list-style-type: none"> <li>Knowledge base in a specific domain</li> <li>Detailed design</li> <li>Validation and implementation</li> </ul>     |
| Implement C++ programs on a personal computer or Arduino microcontroller board using compiler tools.                                  | <ul style="list-style-type: none"> <li>Ability to use appropriate engineering tools and resources</li> </ul>                                              |

## Health and Safety Guidelines

All health and safety rules specific to this course can be found in the lab manual. General health and safety instructions and available health and safety trainings are discussed at:

[Safety Programs - Concordia University](https://www.concordia.ca/campus-life/safety/general-safety.html)

[\(https://www.concordia.ca/campus-life/safety/general-safety.html\)](https://www.concordia.ca/campus-life/safety/general-safety.html)

## ON CAMPUS RESOURCES

Student may find the full list of available resources at:

- [Important Services & Resources - Concordia University \(https://www.concordia.ca/students/services.html\)](https://www.concordia.ca/students/services.html)
- [A-Z Student Services - Concordia University \(https://www.concordia.ca/students/services/a-z.html\)](https://www.concordia.ca/students/services/a-z.html)

While they are included in above mentioned two links, following list includes frequently searched services and resources available for students.

1. [Academic Resources - Concordia University](#)
  - a. [Academic integrity - Concordia University](#)
  - b. [Undergraduate Calendar - Concordia University](#)
  - c. [Graduate Calendar - Concordia University](#) (For Graduate Classes)
  - d. [Academic dates – Undergraduate - Concordia University](#)
  - e. [Important dates – Graduate - Concordia University](#) (For Graduate Classes)
  - f. [Learning services - Concordia University](#)
  - g. [Advocacy - Concordia University](#)
2. [Health & Wellness - Concordia University](#)
  - a. [Access Centre for Students with Disabilities - Concordia University](#)
  - b. [Mental health - Concordia University](#)
3. [Safety & security - Concordia University](#)
  - a. [Security services - Concordia University](#)
  - b. [Environmental Health and Safety - Concordia University](#)
  - c. [Safety Training - Concordia University](#)
4. [International students - Concordia University](#)